

Biogeochemical Processes and the CO₂ system

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**Biogeochemical Processes
and Inorganic Carbon Dynamics**



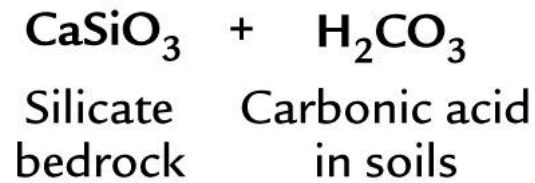
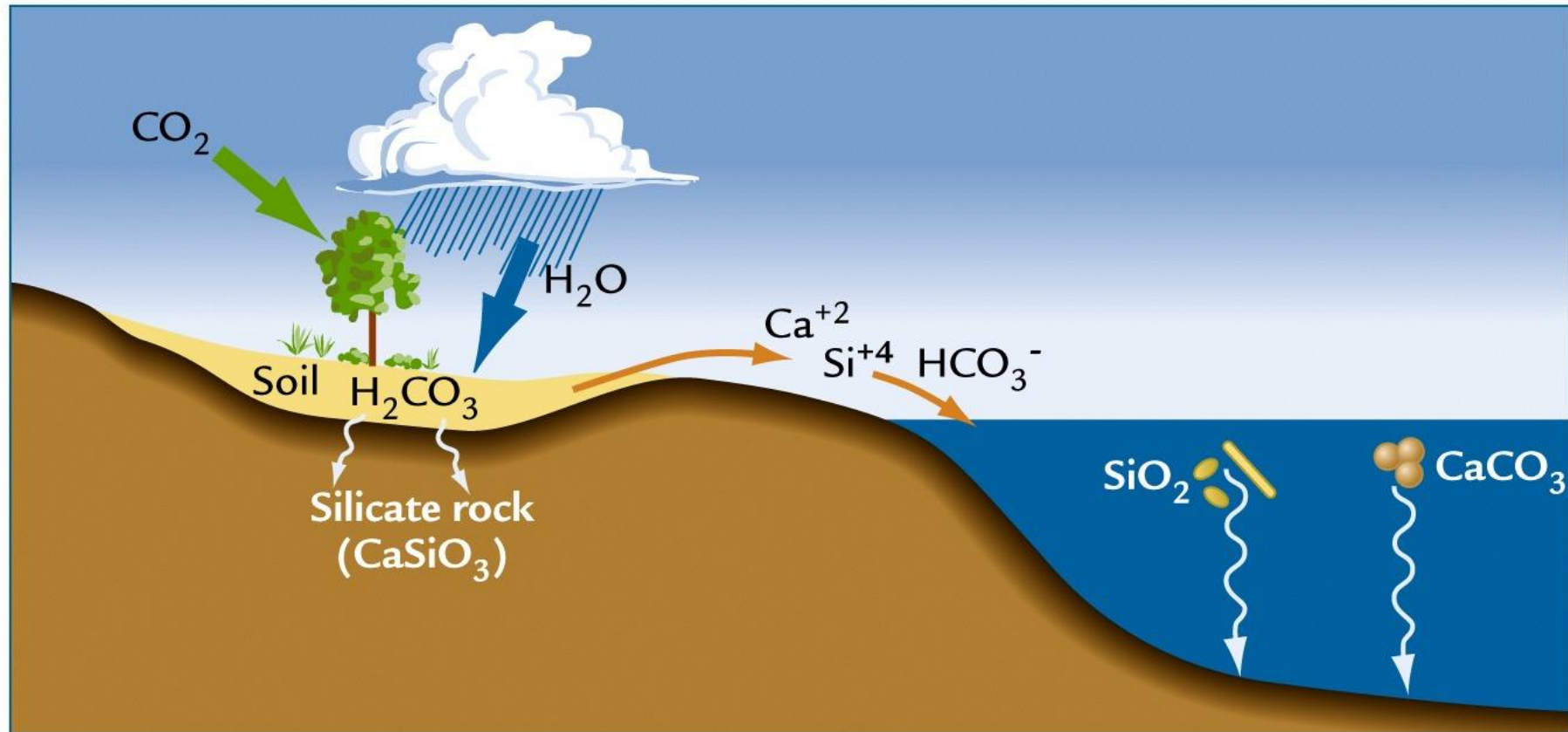
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What you should know after this class

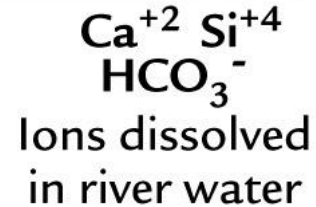
- Long-term C cycle
- The impact of biogeochemical processes on pH



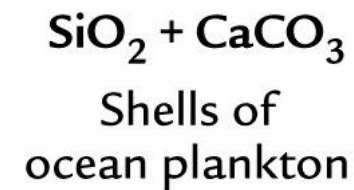
Weathering on continents and mineral formation in ocean



Weathering
on land



Transport
in rivers

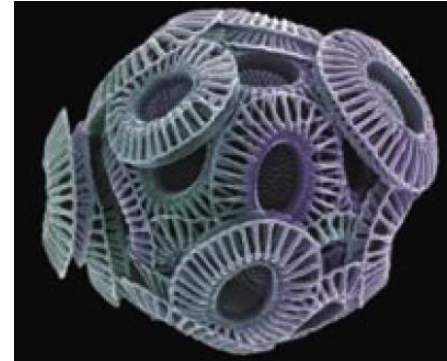


Deposition
in ocean

Long-term C cycle

- Weathering on land balanced by carbonate precipitation in ocean:
 - $\text{CaSiO}_3 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{SiO}_2$
- Rivers transport Ca^{2+} and HCO_3^-
- CaCO_3 precipitates in surface water:
 - $\text{Ca}^{2+} + 2 \text{HCO}_3^- \rightarrow \text{CaCO}_3(\text{s}) + \text{CO}_2 + \text{H}_2\text{O}$
 - **Precipitation releases CO_2**
- Precipitation is biologically controlled

Carbonate minerals in the sea are produced by organisms



CaCO₃ formation: what is right equation?



or



or



or



All reaction schemes for calcium carbonate formation are correct: i.e. they conserve chemical elements and charge

CaCO₃ formation



- All reaction schemes are correct, but
 - (2) forms CO₂
 - (3) generates proton
- Which one is right?
 - All or none of them
- However, they all consume one unit DIC for two units TA
- Dissolution: production of one unit DIC and two units TA

CaCO₃ formation



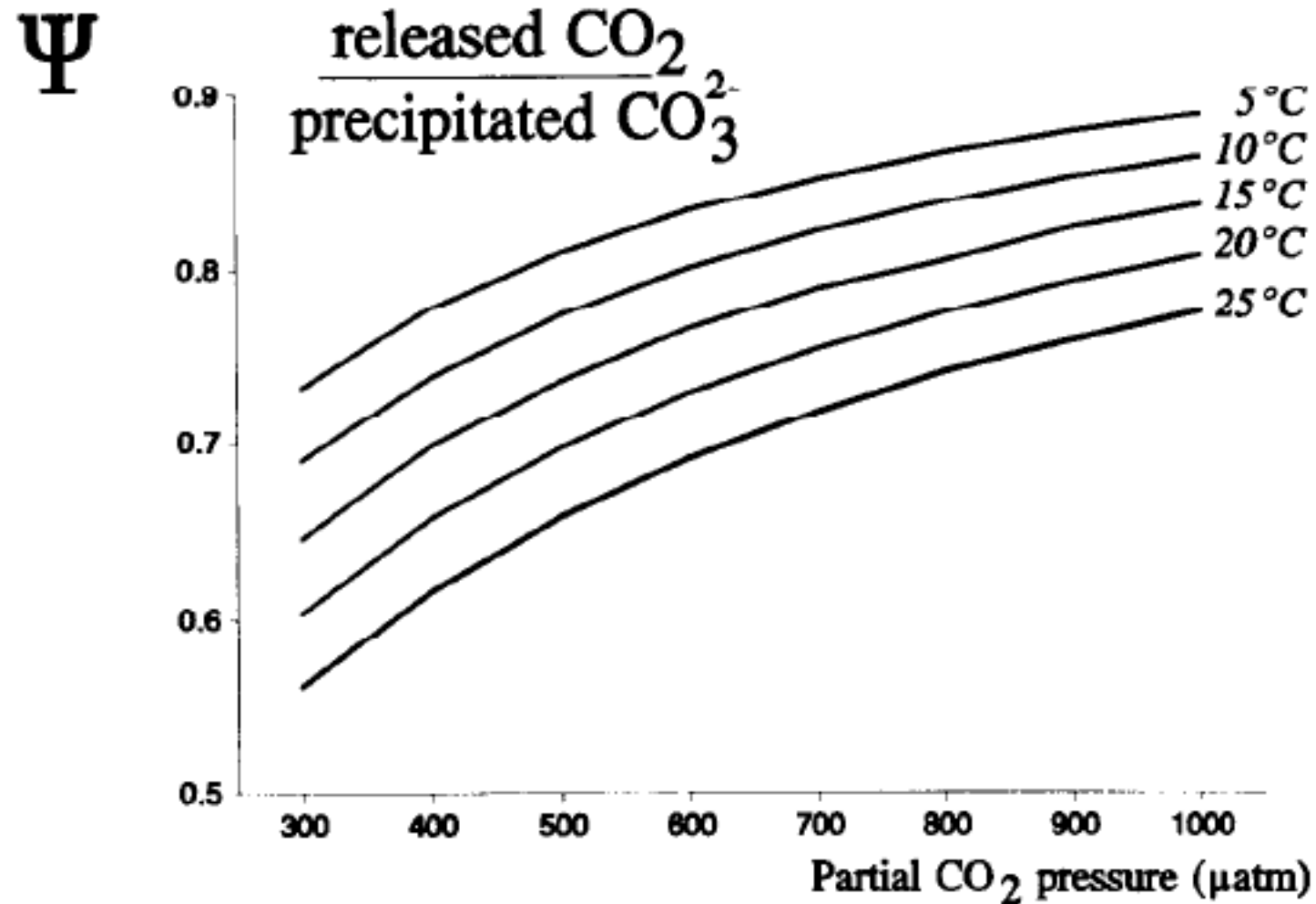
- Relative importance depends on pH
- Whether CO₂ or H⁺ is generated is in principle irrelevant because buffer reactions operate

Frankignoulle Approach

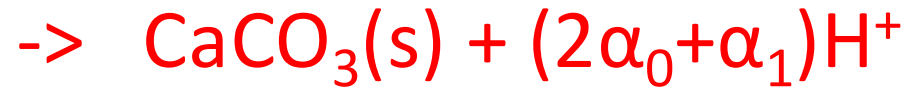


- Frankignoulle et al. (1994) introduced the factor ψ which expresses the amount of CO_2 generated per unit carbonate precipitated.
- ψ is typically 0.6-0.7 for surface water, but increases to 1 under high CO_2 conditions.
- ψ can be calculated from first-principles chemistry and is consistent with observations

More CO₂ released in future



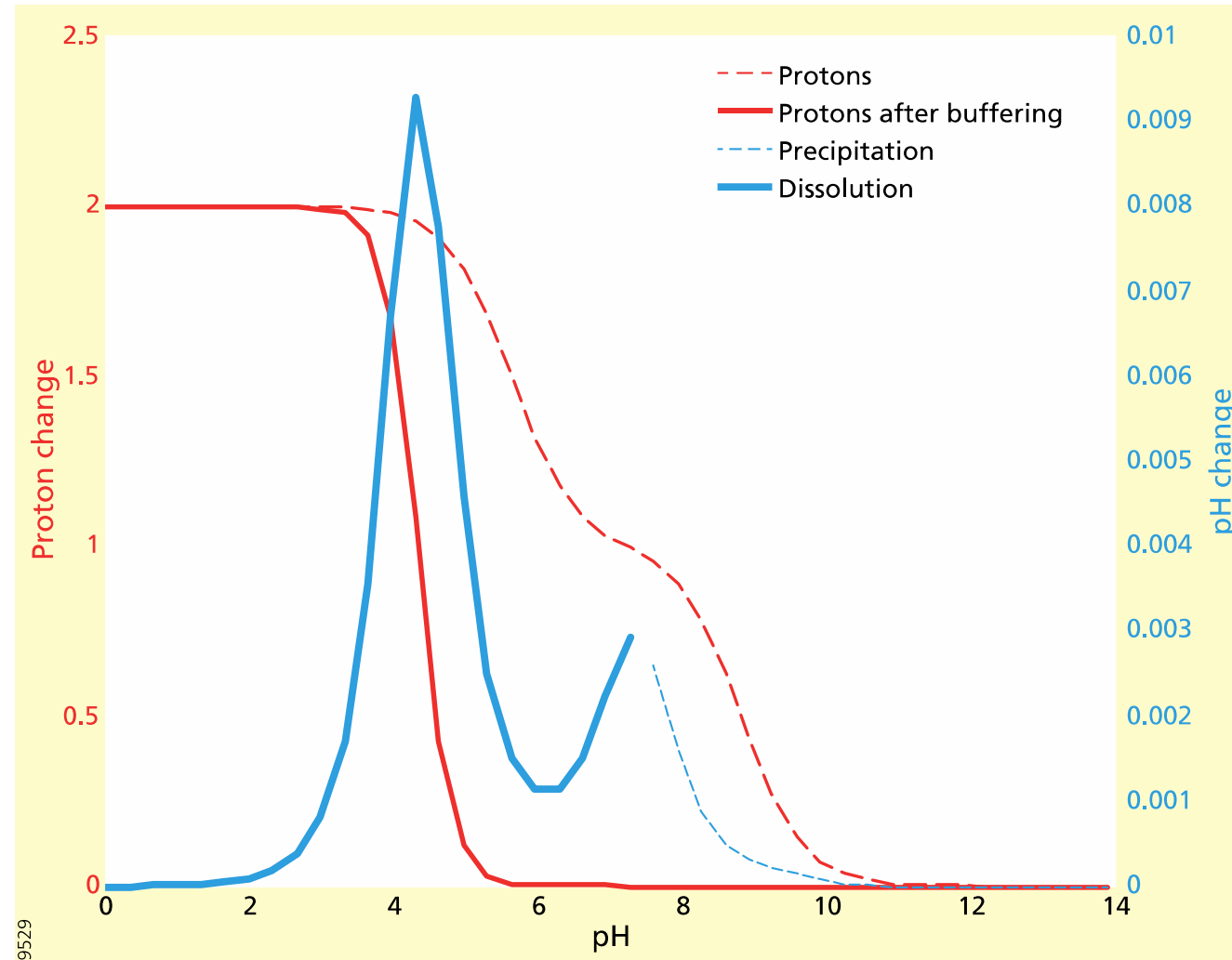
Hofmann Generalization



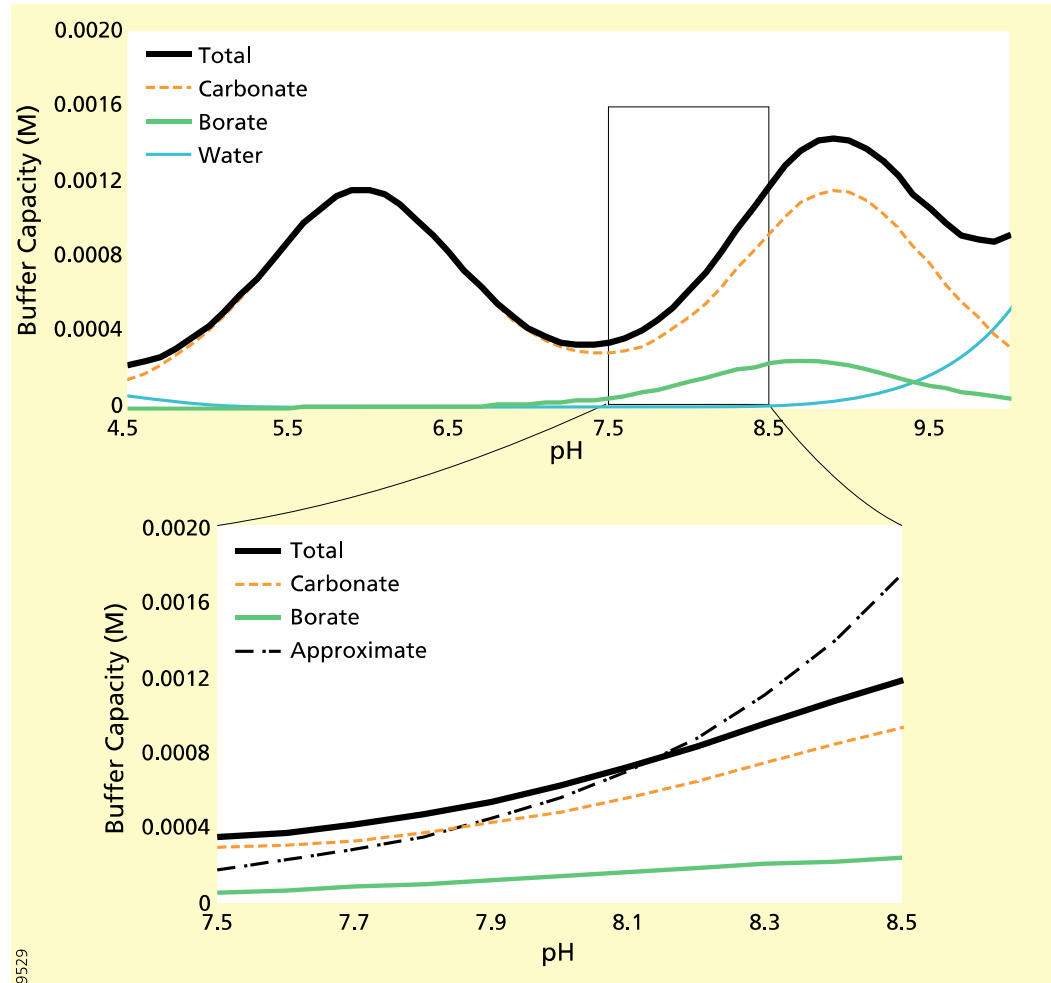
where $\alpha_0 + \alpha_1 + \alpha_2 = 1$ for any reaction written for calcium carbonate precipitation.

Where α_0 , α_1 , α_2 are the well-known ionization fractions: e.g. $\alpha_0 = \text{CO}_2/\text{DIC}$, $\alpha_1 = \text{HCO}_3^-/\text{DIC}$, etc

H⁺ production, buffering and change in pH

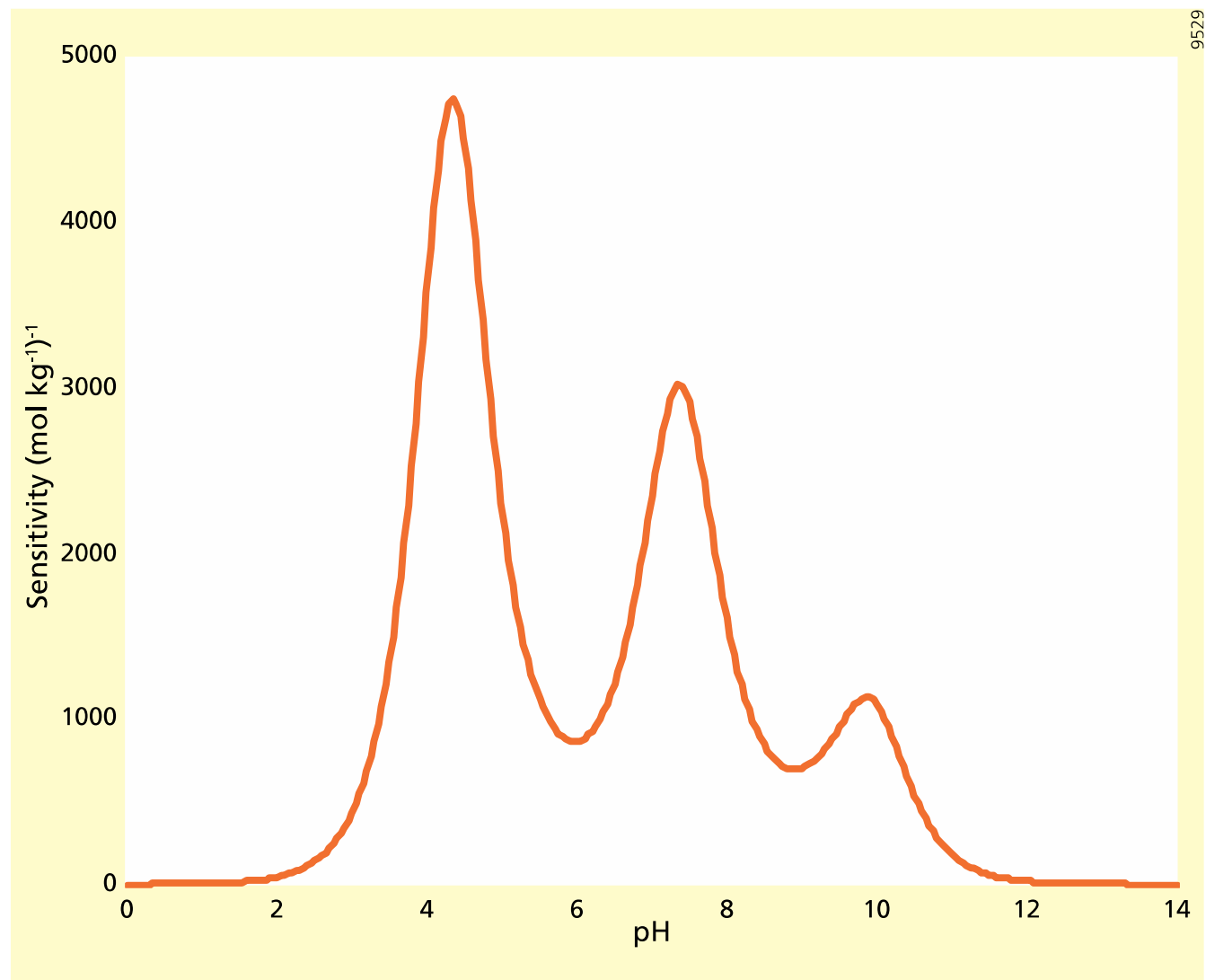


Sensitivity and Buffering

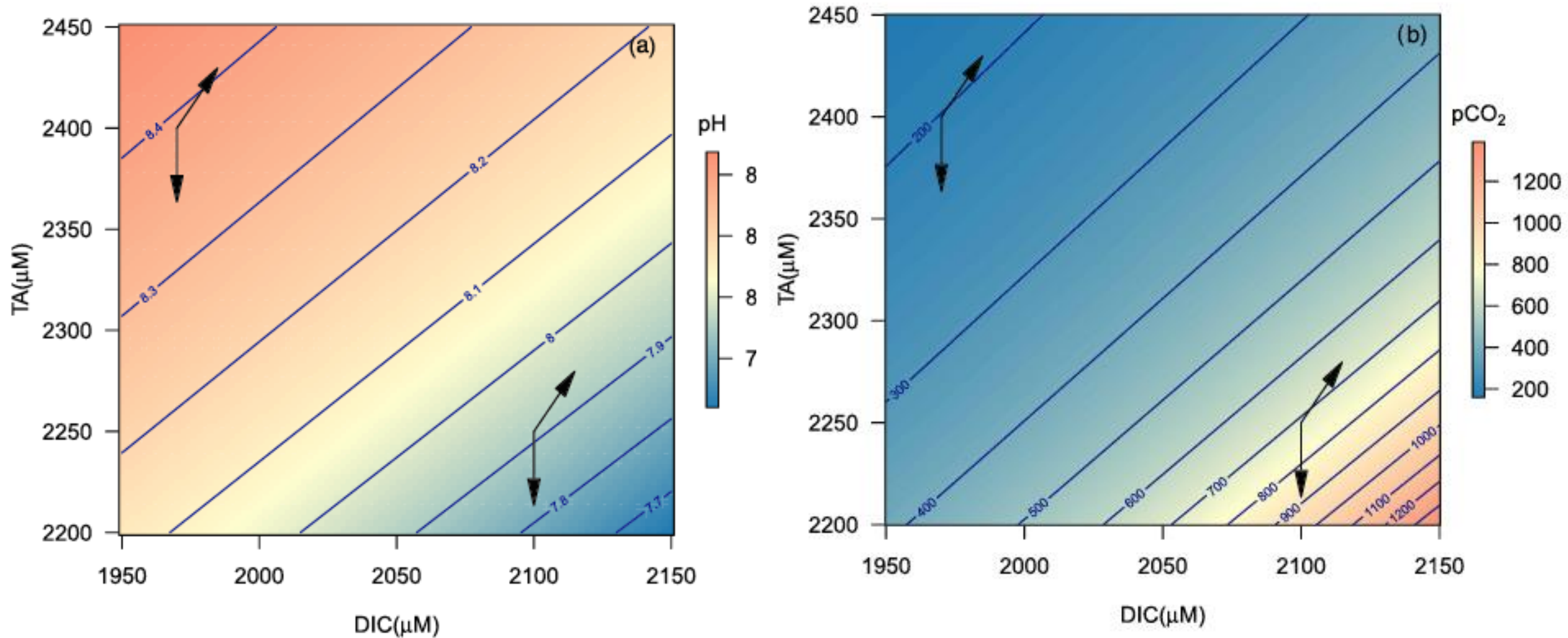


- Sensitivity expresses the change in X given a change in Y
- Buffering capacity is a measure for attenuation of changes (reverse of sensitivity). Carbonate system governs buffering

Sensitivity functions

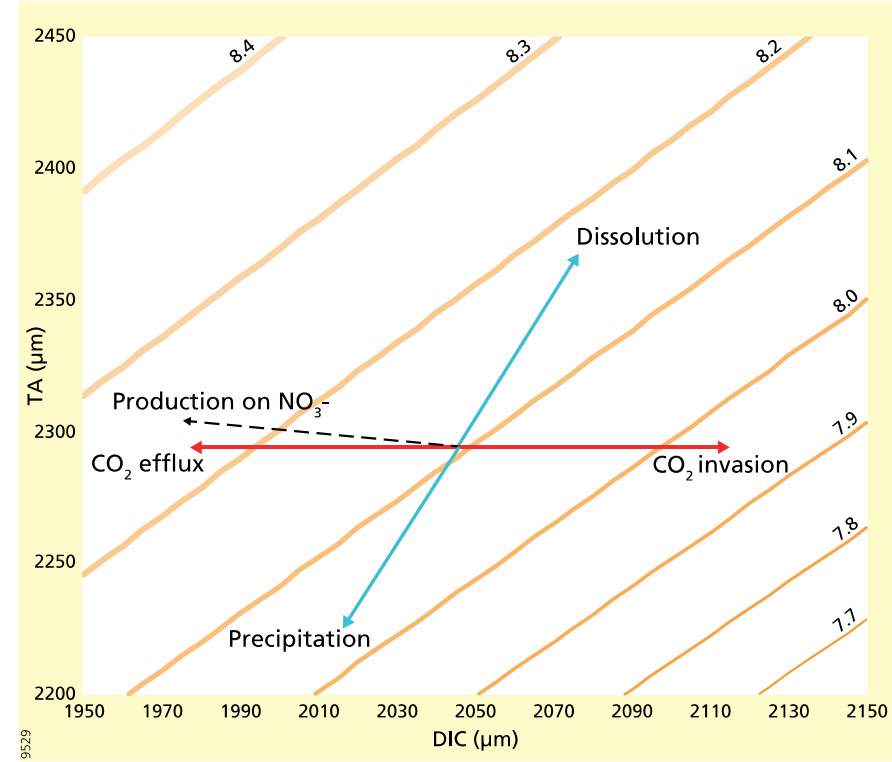
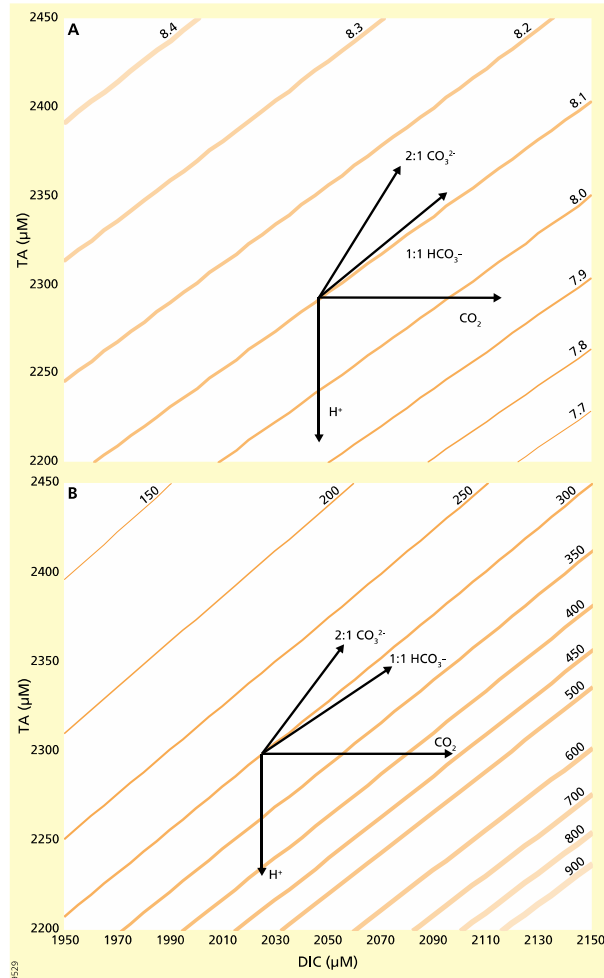


Biogeochemical process and pH



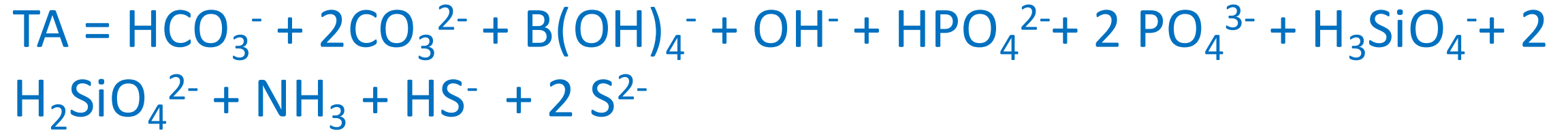
Same vector: different effect because of sensitivity/buffering

Biogeochemical process and pH: Graphical Approach



Deffeyes (1965) diagram

Impact Biogeochemical Process on TA: Denitrification

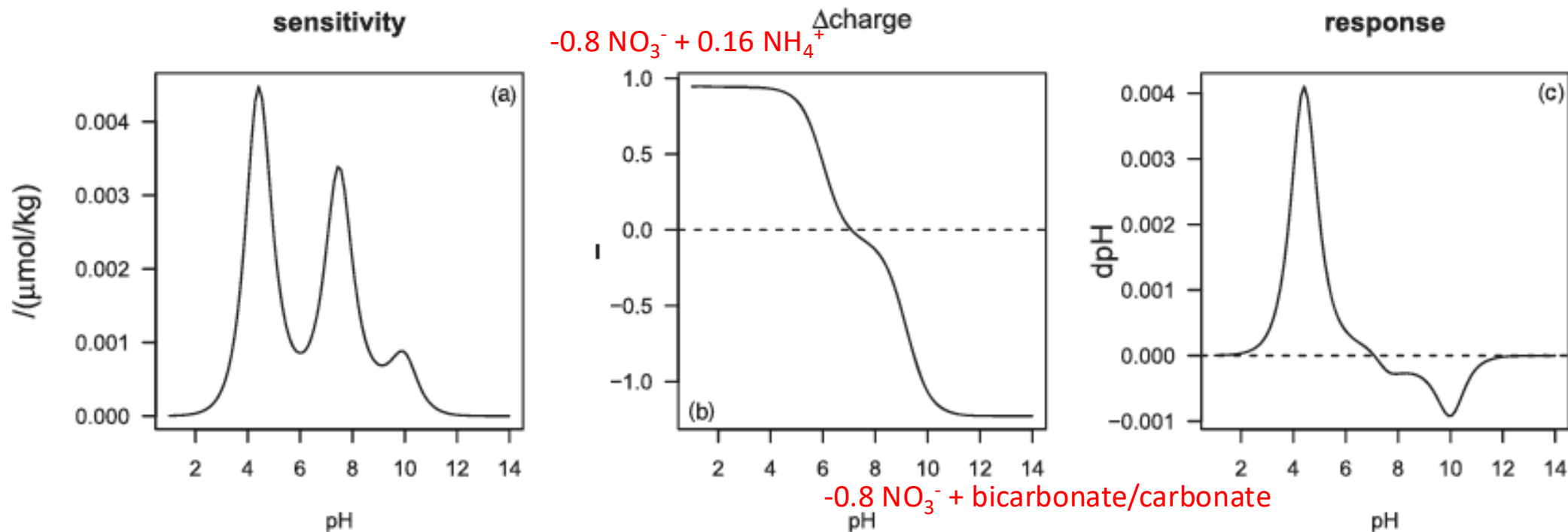


Change in TA: $0.8 + n - p$:

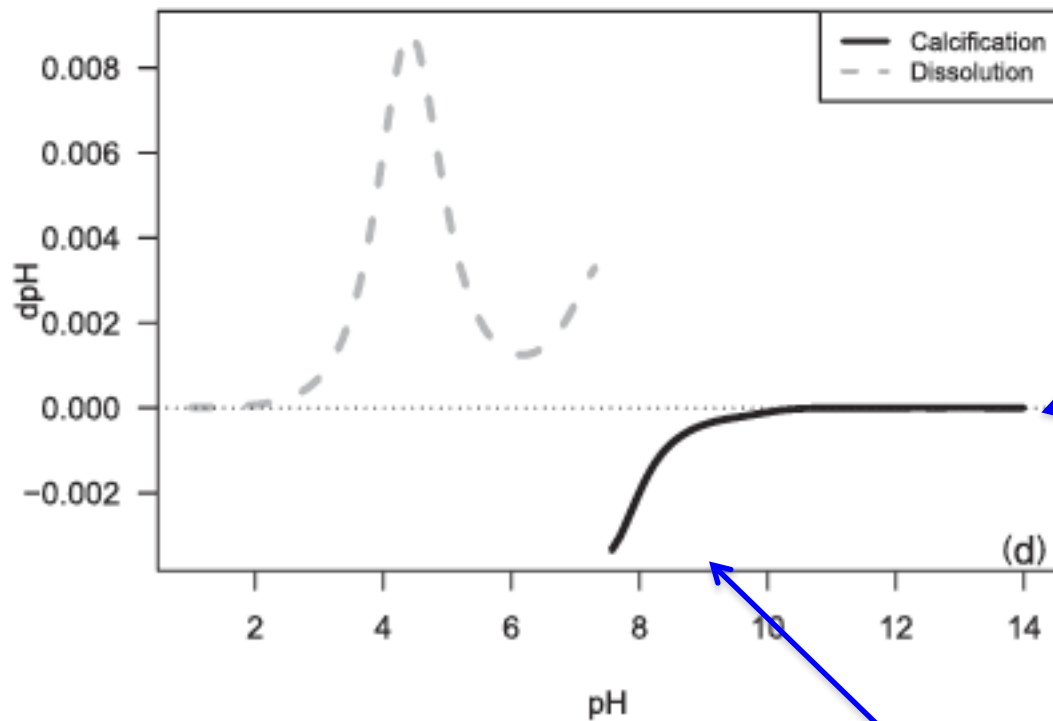
- 0.8HNO_3 consumed - -0.8
- $n\text{NH}_3$ produced: n
- $p\text{H}_3\text{PO}_4$ produced: $-p$

Biogeochemical process and pH: Theoretical Approach

$$\text{dpH} = \Delta \text{ charge} * \text{sensitivity} * \text{intensity process}$$

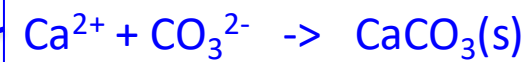


ΔpH depends on pH: CaCO_3 production & dissolution



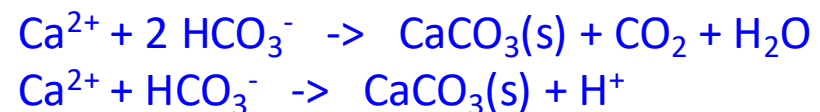
Carbonate dissolution

- consumes protons
- releases alkalinity
- Increases pH



Carbonate precipitation

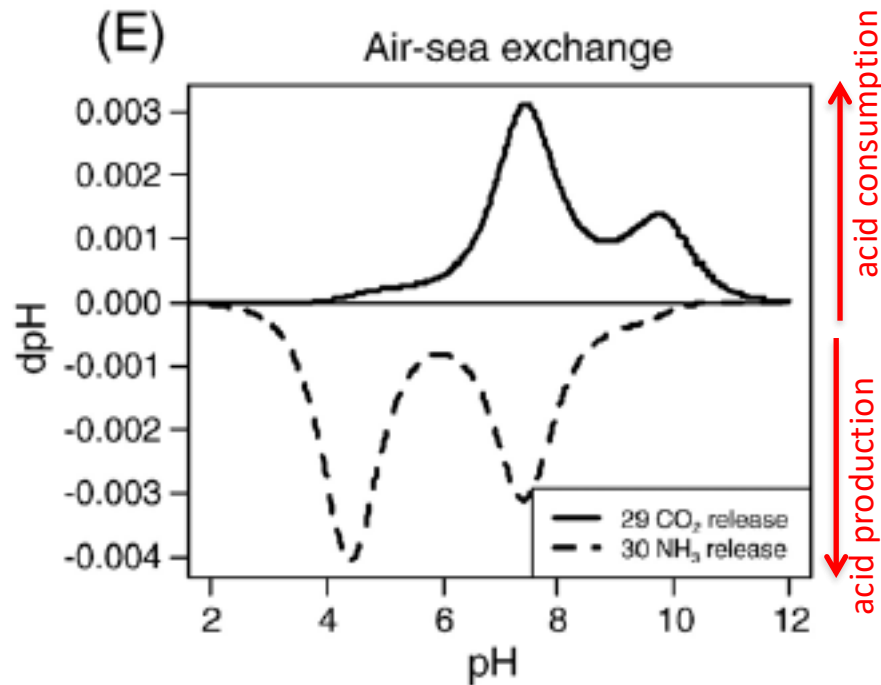
- produces protons
- consumes alkalinity
- decreases pH



CO₂ exchange with atmosphere

- CO₂ exchange only impacts DIC not TA
 - But as a consequence via CO₂ system, the pH
- CO₂ outgassing: increase in pH
- CO₂ invasion: decrease in pH (ocean acidification)

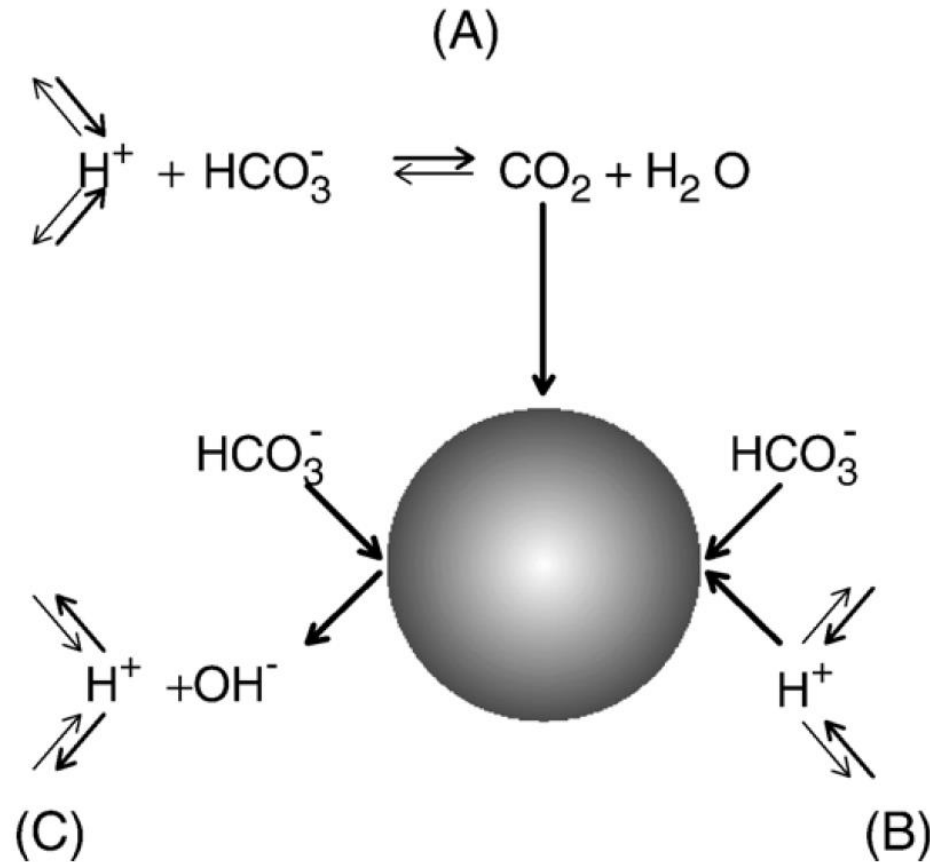
ΔpH depends on pH: CO_2 release



CO_2 outgassing

- consumes protons
- Increases pH

DIC uptake by phytoplankton



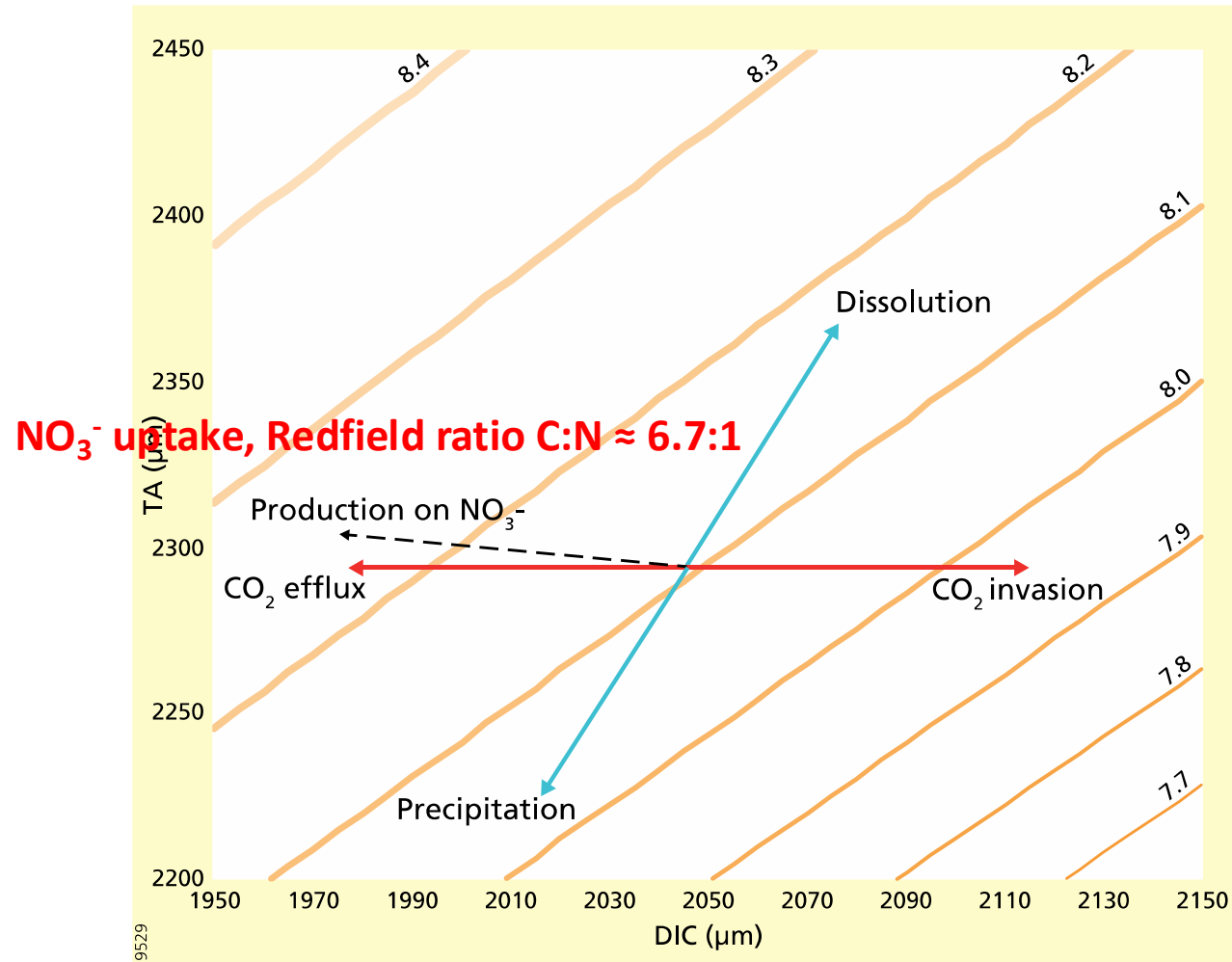
Whether
 CO_2 (A)
or HCO_3^- (B, C)
is taken up,
the pH increases
with DIC
consumption by
phytoplankton

Fig. 3. Schematic representation of DIC uptake whilst conserving the electroneutrality of the solution. A. Uptake of (uncharged) CO_2 . B. Uptake of HCO_3^- , negative charge balanced by uptake of H^+ . C. Uptake of HCO_3^- , negative charge balanced by release of OH^- .

Nutrient-proton compensation

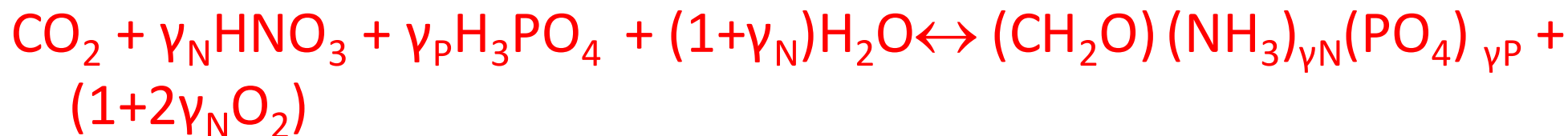
- Ionic nutrients (NH_4^+ , NO_3^- , PO_4^{3-} , HCO_3^-) are taken up by phytoplankton
- To maintain electroneutrality, H^+ (or OH^-) is exchanged
 - NO_3^- uptake implies H^+ uptake
 - NH_4^+ uptake implies H^+ release
 - PO_4^{3-} or HPO_4^{2-} uptake implies H^+ uptake

Impact of primary production



Nutrient-proton compensation

- For generalised Redfield organic matter we then arrive for photosynthesis based on nitrate at:



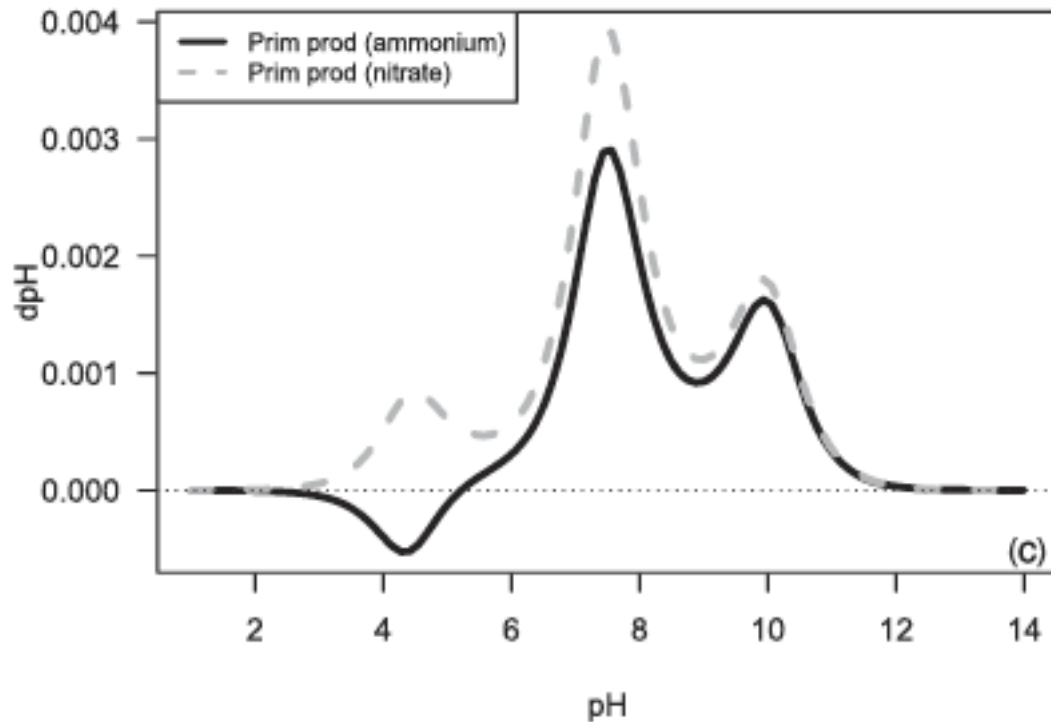
$$\Delta\text{TA} = \gamma_N + \gamma_P = (16/106 + 1/106) = 0.16 \text{ vs. } \Delta\text{DIC} = 1$$

- Based on ammonium at:



$$\Delta\text{TA} = -\gamma_N + \gamma_P = (-16/106 + 1/106) = -0.14 \text{ vs. } \Delta\text{DIC} = 1$$

ΔpH depends on pH: photosynthesis



CO_2 consumption:

- increases pH because of re-equilibration

HCO_3^- consumption:

- Increase pH because of H^+ uptake for charge balance

NH_4^+ uptake:

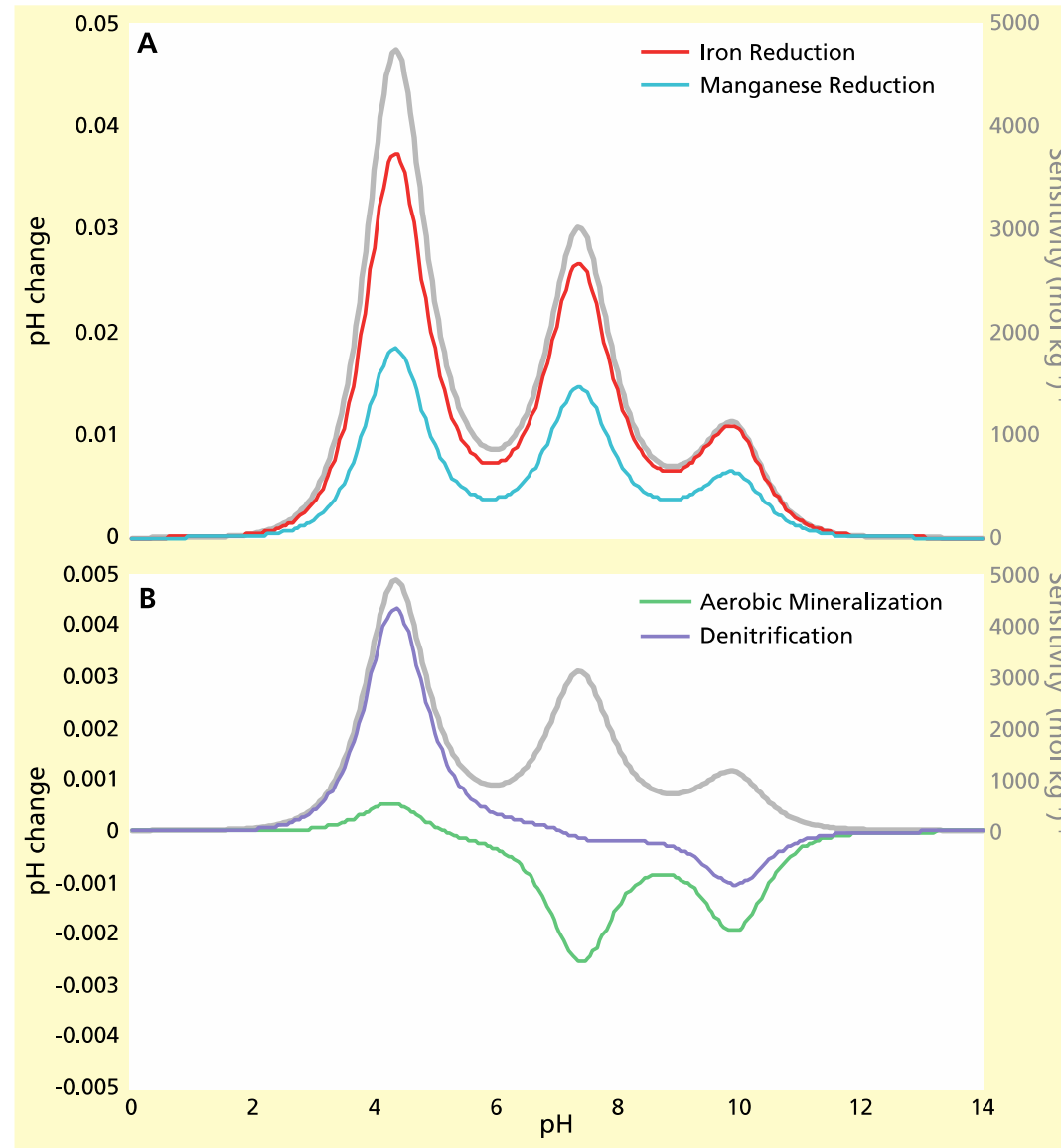
- Decrease in pH because of proton release for charge balance

NO_3^- uptake:

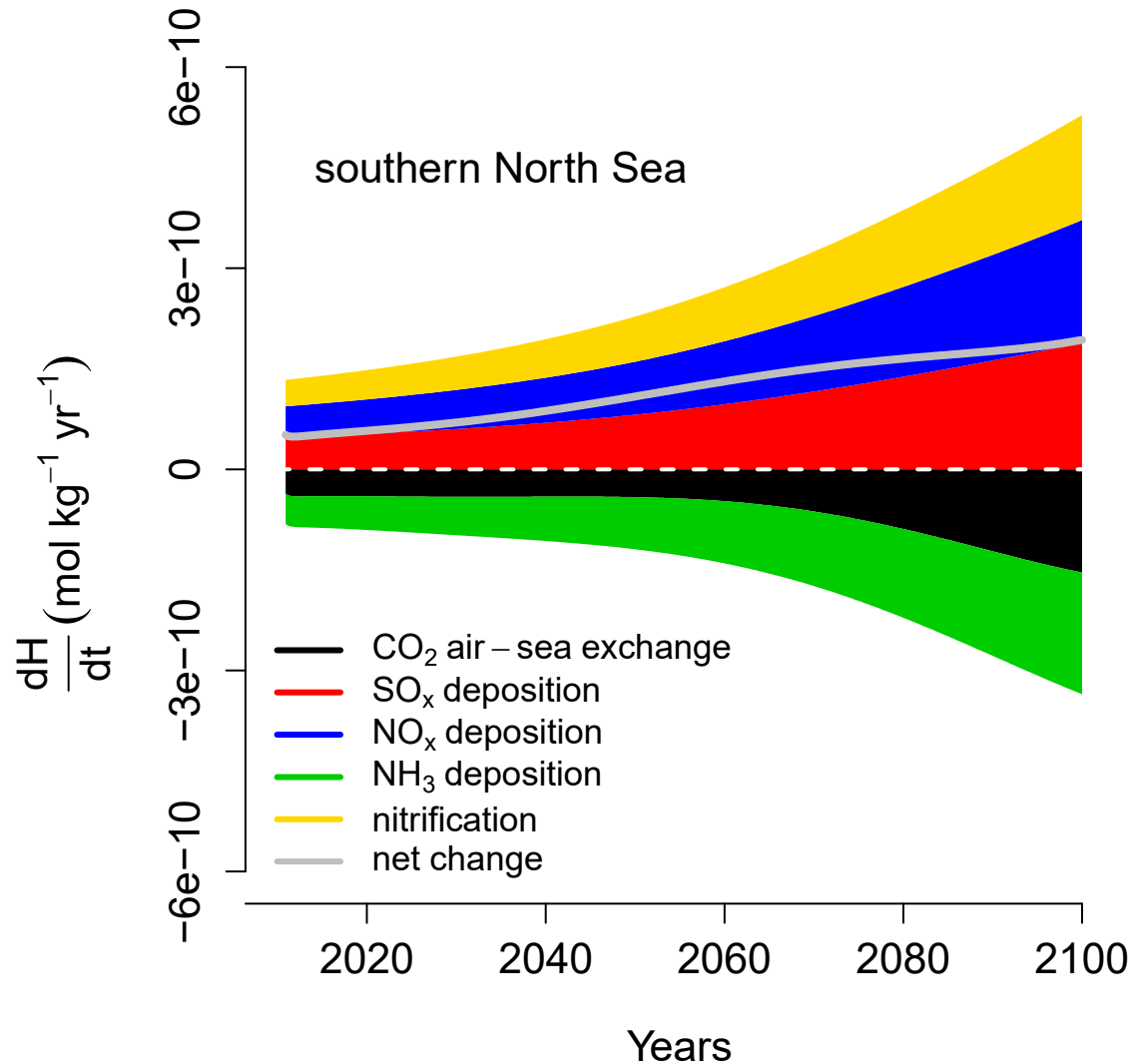
- Increase in pH because of proton uptake for charge balance

acid consumption
acid production

pH change and respiration



pH evolution North Sea

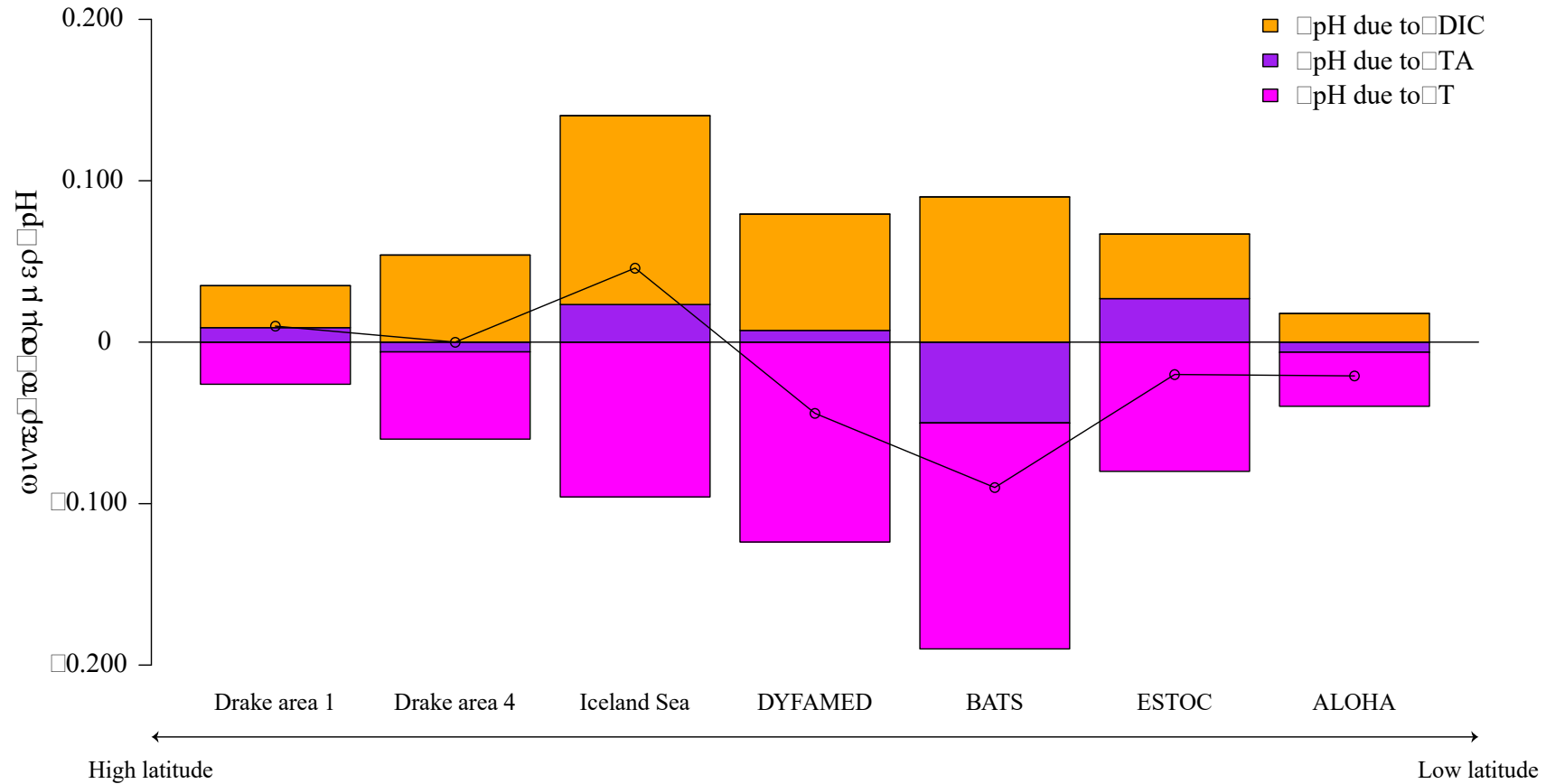


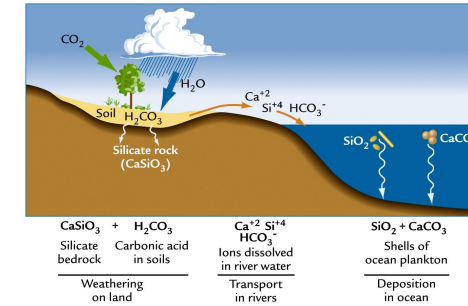
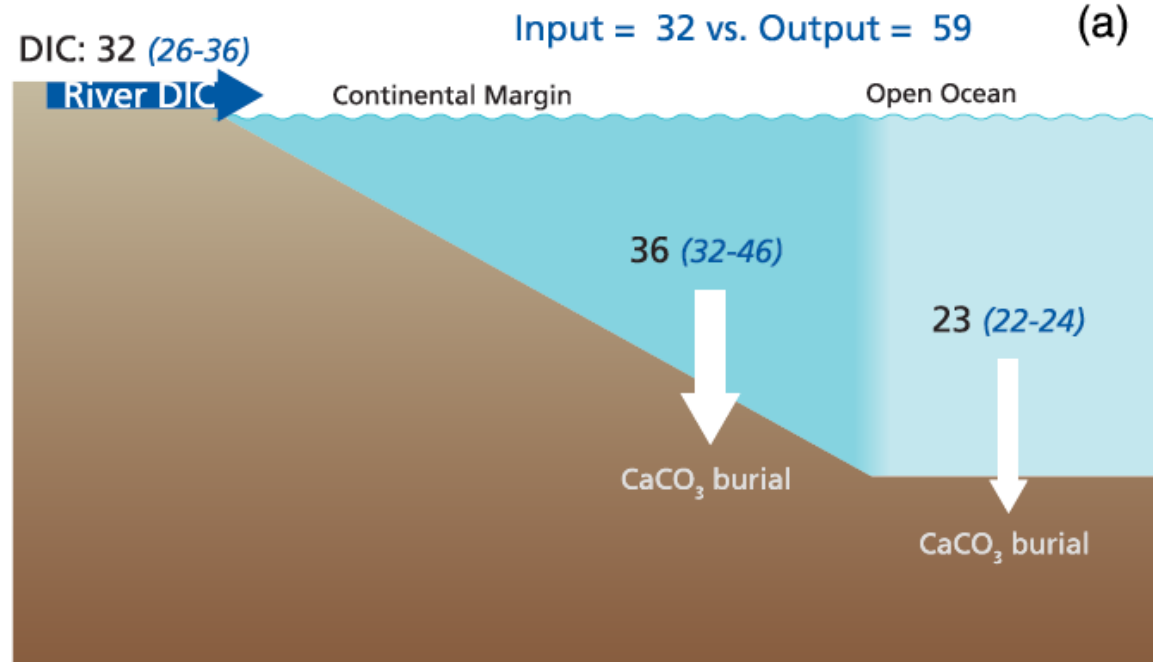
acid production
acid consumption

With fixed atmospheric deposition and increasing atmospheric carbon dioxide, there will be more acid production/consumption because of decrease in buffering:

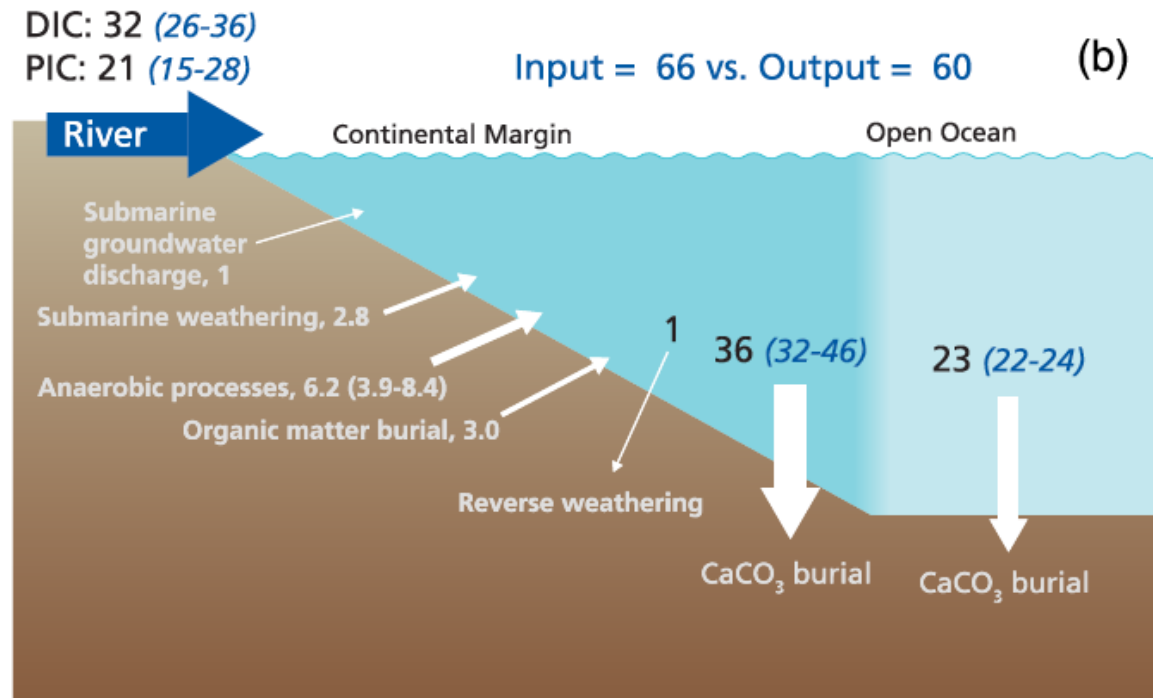
- ocean becomes about 4 times more vulnerable to acid deposition by end of century

What governs seasonality in pH





Traditional Alkalinity Balance Ocean



Alkalinity Balance of biogeochemically active ocean